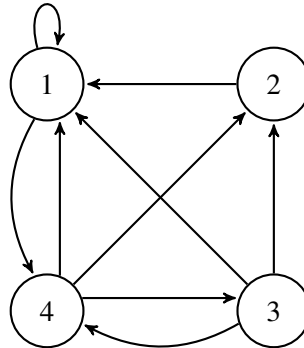


1. Adjacency and Connectivity Matrices

Consider the directed graph below. An arrow from node j to node i means that page j links to page i .



Graph B

- (a) Find the adjacency matrix A_B for graph B. Recall that for a directed graph,

$$(A_B)_{ij} = \begin{cases} 1 & \text{if there is an edge from node } j \text{ to node } i, \\ 0 & \text{otherwise.} \end{cases}$$

Answer: With rows and columns ordered 1, 2, 3, 4,

$$A_B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

Each column lists the destinations of links from that node. The self-loop at node 1 gives $(A_B)_{11} = 1$.

- (b) Each page splits one vote equally among the pages it links to, including itself if it has a self-loop. The connectivity matrix entry $(C_B)_{ij}$ is the share of page j 's vote that goes to page i .

Find C_B by normalizing the columns of A_B . Check that each column sums to 1, and explain why this follows from the voting rule. A nonnegative matrix whose columns each sum to 1 is called *column stochastic*.

Answer: The numbers of outgoing links are 2, 1, 3, 3, respectively. Dividing each column by its number of outgoing links gives

$$C_B = \begin{bmatrix} \frac{1}{2} & 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{3} & 0 \end{bmatrix}.$$

The column sums are $\frac{1}{2} + \frac{1}{2} = 1$, 1 , $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, and $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$. Each column distributes exactly one page's full vote, so no vote is lost or created. For example, page 2 sends its entire vote to page 1: $(C_B)_{12} = 1$.

2. Mechanical Least Squares

We observe the following data and would like to fit a line:

$$\begin{array}{c|cccc} a_i & 2 & 4 & 6 & 8 \\ \hline b_i & 2 & 6 & 7 & 8 \end{array} \quad \mathbf{a} = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

First consider the model $b_i \approx sa_i$, which forces the fitted line to pass through the origin. The slope s is the only unknown parameter.

- (a) Write the squared error $L(s) = \sum_{i=1}^4 (b_i - sa_i)^2$ as a quadratic in s . Differentiate to find the best slope $\hat{s}$, and justify that it gives a minimum. Compute the minimum squared error.

Answer: Expanding gives

$$L(s) = (2 - 2s)^2 + (6 - 4s)^2 + (7 - 6s)^2 + (8 - 8s)^2 = 120s^2 - 268s + 153.$$

Thus $L'(s) = 240s - 268 = 0$ gives $\hat{s} = 67/60$. Since $L''(s) = 240 > 0$, this is the unique global minimum. Substitution gives $L(67/60) = 101/30 \approx 3.367$.

- (b) Express $L'(s)$ using $\mathbf{a}$ and the residual $\boldsymbol{\varepsilon}(s) = \mathbf{b} - s\mathbf{a}$. What does $L'(\hat{s}) = 0$ say geometrically? Compute $\boldsymbol{\varepsilon}(\hat{s})$ and verify this condition directly.

Answer: Differentiating the sum before expanding gives

$$L'(s) = -2 \sum_{i=1}^4 a_i(b_i - sa_i) = -2\mathbf{a}^\top \boldsymbol{\varepsilon}(s).$$

At the minimum, $\mathbf{a}^\top \boldsymbol{\varepsilon}(\hat{s}) = 0$, so the residual is perpendicular to the line of possible predictions $\text{span}\{\mathbf{a}\}$. In particular,

$$\boldsymbol{\varepsilon}(\hat{s}) = \begin{bmatrix} -7/30 \\ 23/15 \\ 3/10 \\ -14/15 \end{bmatrix}, \quad \mathbf{a}^\top \boldsymbol{\varepsilon}(\hat{s}) = \frac{-14 + 184 + 54 - 224}{30} = 0.$$

Here the prediction is a vector in $\mathbb{R}^4$; the fitted line through the plotted data is a separate picture in the (a, b) plane.

(c) Now fit the model $b_i \approx sa_i + c$, allowing an intercept c . Write the parameter vector as

$$\mathbf{x} = \begin{bmatrix} s \\ c \end{bmatrix}.$$

Construct the model matrix A so that the matrix-vector product $A\mathbf{x}$ contains all four predictions. Use the normal equations to find $\hat{s}$ and $\hat{c}$. Compute the squared error and compare it with part (a).

Answer: The column of ones allows the same intercept to be added to every prediction:

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \\ 6 & 1 \\ 8 & 1 \end{bmatrix}, \quad A^\top A = \begin{bmatrix} 120 & 20 \\ 20 & 4 \end{bmatrix}, \quad A^\top \mathbf{b} = \begin{bmatrix} 134 \\ 23 \end{bmatrix}.$$

Solving $120\hat{s} + 20\hat{c} = 134$ and $20\hat{s} + 4\hat{c} = 23$ gives $\hat{s} = 19/20$ and $\hat{c} = 1$. The predictions and residual are

$$\hat{\mathbf{b}} = \begin{bmatrix} 29/10 \\ 24/5 \\ 67/10 \\ 43/5 \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} -9/10 \\ 6/5 \\ 3/10 \\ -3/5 \end{bmatrix}.$$

The squared error is $\|\boldsymbol{\varepsilon}\|^2 = 27/10 = 2.7$, which is smaller than $101/30$.